

Predicting House Prices

Regression Analyses

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Introduction



- House prices can be difficult to predict due to the number of potentially significant variables
- Prices have been increasing for decades along with interest rates on mortgage loans
- Some variables are not easily quantified or accounted for without leading to an extremely complicated model
 - Variables representing economic state
 - Variables representing proximity to specific landmarks or locations

Why?

Rationale

- House pricing affects everyone, directly or indirectly
- Can be an indicator of the quality of life
- Developing a method of predicting house price can be useful for several groups
 - Homeowners determining the value of their property
 - Prospective homeowners deciding where to live
 - Economists studying important facets of day-to-day life
 - Politicians developing or adjusting housing policies

Background Research



Economic Perspective

- Supply and demand principles
- Employment and interest rates influencing house price
- Proximity to landmarks such as schools, city hubs, etc.

Mathematical Perspective

- Studies concluded statistically significant factors:
 - House quality
 - Area, or square footage
- Typically use regression techniques, most often linear regression

Proposal

Develop a model using the Ames housing data set which most accurately and reliably predicts the house prices

- Machine Learning Algorithms
 - Linear Regression
 - Ridge Regression
 - Gradient Boost Regression
- Splitting Dataset
 - Training models
 - Testing models



Outline

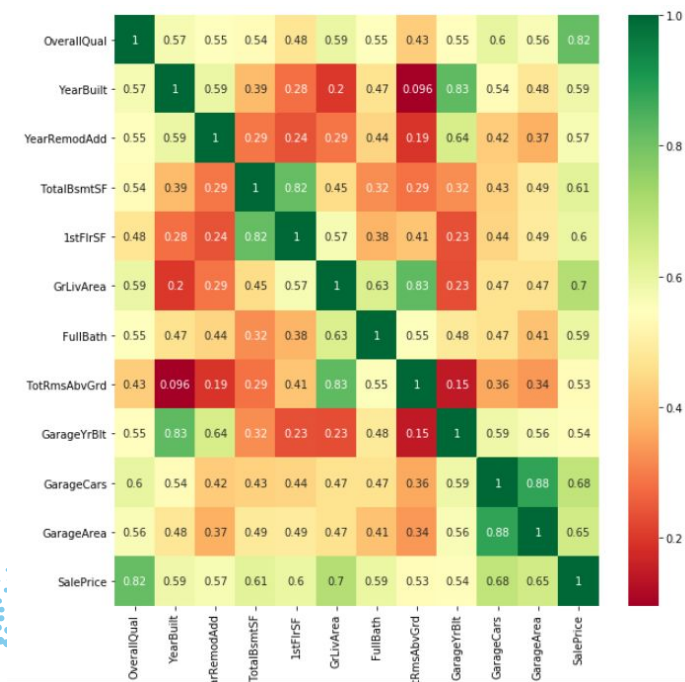
- Exploratory data analysis
 - Data clearing/cleaning
 - Transformations
 - Training and Test data split
- Three regression techniques to predict prices for input houses
 - Multiple Linear Regression
 - Bayesian Ridge Regression
 - Gradient Boost Regression
- Assess models through error statistics
- Compare results, draw conclusions

Dataset

- Ames, Iowa from 2006 to 2010
 - Alternative to Boston Housing Data
 - Ethical issues
 - Applications to other locations and other time periods may be limited
- 2930 observations, 80 explanatory variables
 - 23 nominal, 23 ordinal, 14 discrete, 20 continuous
 - House type, item quality, house items, and area dimensions respectively
 - All variables assured to be understandable
 - No special knowledge needed
 - No previous calculations needed



Exploratory Data Analysis



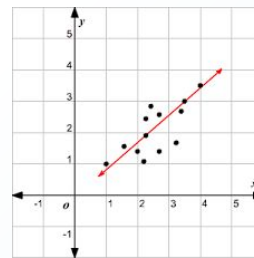
- 80 explanatory variables
 - High chance for multicollinearity
 - Variables with missing data
 - Removed if > 50% missing
 - Highly skewed data (sale price)
 - Log-transformed
- Correlation matrix
 - Address multicollinearity by acknowledging explanatory variables which are “redundant” of each other
 - Initial guesses on most important factors:
 - Overall quality
 - Garage area
 - Garage capacity

Data Cleaning

- Log transformation for variables that distribution is skewed
- Fill in mean for missing values for quantitative variables
- Fill in mode for missing values for qualitative variables
- Dropped any variables that are missing large portion of



Multiple Linear Regression



- Estimate coefficients for linear combination of explanatory variables
 - Fitted equation used for predictions: $\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$
- Objective (OLS): $\min \sum_{i=1}^n (\hat{f}(x_i) - y_i)^2$
- Assumptions
 - Linear relationship between dependent / independent variables
 - Multivariate normality
 - Limited multicollinearity
 - No autocorrelation between errors
 - Homoscedasticity
- Assessed with root mean squared error
- Possible chance of underfitting due to complexity of data (low variance but high bias)

MLR Model

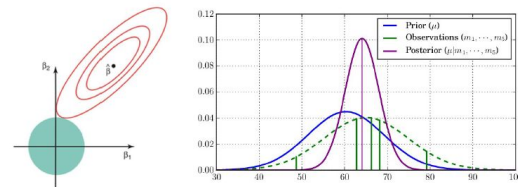
- We used 10 fold cross validation to evaluate the model
- The result came out to be Inf. values since the model was poor to fit the data



MLR Results

- Root mean square error: INF
 - Predicted values very far off from actual values, computer displays massive number as infinity
- Standard Deviation: N/A
 - Method simply does not work with the data
 - RMSE is infinite, residual standard deviation incalculable
- Extreme underfitting, patterns not captured by the model
 - Low variance in fitted models
 - High bias in predicted values
- Multicollinearity a possible factor in poor performance
 - Reduced parameter estimate reliability
 - Inaccurate predictions made
 - Large errors

Ridge Regression



- Similar to Multiple Linear regression, used when there is significant multicollinearity
- L2-regularization
 - Decrease weight vectors without having them reach 0 (weight decay)
- Objective:
$$\min \left(\sum_{i=1}^n ((x^{(i)})^T w - y^{(i)})^2 + \lambda ||w||^2 \right)$$
 - Lambda controls magnitude of regularization
- Variables standardized
- Assessed with root mean squared error

Ridge Regression Model

- We used 10 fold cross validation to evaluate the ridge regression model
- The result seem to be fitting the data well. We took the average of the 10 results.

10 fold cross validation results :

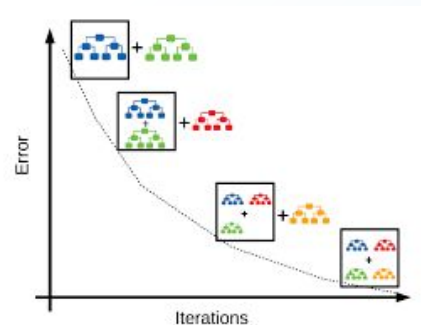
```
array([1.13412011, 1.11207253, 1.12869549, 1.17434761, 1.16022904,  
       1.16631068, 1.13454088, 1.10330329, 1.21536545, 1.1351858 ])
```

Ridge Regression Results

- Root mean squared error: 1.1464
 - Very low considering volume of data set
 - Potential overfitting concern
- Standard deviation: 0.0206
 - Low variance of residuals
 - Most residuals close to the mean, assumed to be 0
 - Strong indicator of high performance
- Performed significantly better than linear regression, regularization term crucial

Gradient Boost Regression

- Ensemble of simpler models
 - Model improves itself by adding additional estimators (decision trees)
- Minimizes loss function using gradient boost algorithm
- $\mathbf{w}^{k+1} = \mathbf{w}^k - \alpha_k \nabla \ell(\mathbf{w}^k)$
- Possible constraints
 - Limit size and shape of trees
 - Specific weights for each tree
 - Random forest (reduce multicollinearity)
 - Penalized learning (regularization)
- Assessed with root mean squared error



Gradient Boost Regression Model

- We used 10 fold cross validation to evaluate Gradient Boost Regression
- The result seem to be fitting the data well. We took the average of the 10 results.

10 fold cross validation results :

```
array([1.13881084, 1.10835475, 1.11856053, 1.16764122, 1.16587377,  
       1.11194871, 1.14501052, 1.11718538, 1.11967562, 1.14695495]),
```

Gradient Boost Results

- Root mean squared error: 1.1341
 - Very low, fits data extremely well
 - Overfitting concern
- Standard deviation: 0.0314
 - Low variance of residuals
 - Most residuals close to mean, which is assumed to be 0
 - Further indicates high performance
- Similar performance to ridge regression
 - Slightly lower error
 - Higher standard deviation

Discussion



- Linear regression not feasible with selected data set
 - Extreme underfitting and bias
 - Further data manipulation needed
 - Elimination of variables with large VIFs
 - Limit total number of variables
- Ridge Regression and Gradient boost performed very similarly
 - Reasonable means of predicting house prices
 - Found overall quality to be the most significant factor
 - Household items, garage capacity, also important
 - Possible overfitting concerns
 - Ridge: regularization forces weights to be small, but less explanatory variables could be safer
 - Gradient: implement more tree constraints
- Ridge regression may be selected as optimal model
 - Comparatively low RMSE to Gradient Boost
 - Lower standard deviation of residuals
- Results predominantly inline with previous studies conducted on house prices

Conclusions



- Linear regression may be possible to model the data set, though would require significant modifications to the data and methodology
- Ridge Regression and Gradient Boost performed so similarly that the two models are interchangeable
 - Gradient Boost had lower RMSE but Ridge Regression had lower SD
 - Distance between the two RMSE was marginally larger than the distance between the two SD
- Possible improvements
 - Experimenting with other modeling methods
- Most significant variables depended on the properties of the houses while location was found to be less significant

Sources

- Dataset explanation:
 - <http://jse.amstat.org/v19n3/decock.pdf>
- Past Analysis on Boston Dataset:
 - <https://www.hindawi.com/journals/sp/2021/7678931/>
- Economic Sources:
 - <https://www.economicshelp.org/blog/377/housing/factors-that-affect-the-housing-market/>
 - <https://www.opendoor.com/w/blog/factors-that-influence-home-value>